Junjie Yu

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Education

M.S. in Mathematics — Southern University of Science and Technology (SUSTech), Shenzhen, China *2021 – 2023*

Research focus: Optimization / Inverse Problem Reconstruction

B.S. in Mathematics and Applied Mathematics — Hunan Normal University, Changsha, China *2015 – 2019*

Work Experience

Image Algorithm Engineer — Shenzhen Wisonic Medical Technology Co., Ltd. *Jun 2023 – Apr 2025*

System Algorithm Engineer — Krinwave Technology (Shenzhen) Co., Ltd. *May 2025 – Jun 2026*

Key Projects:

1. **Anisotropic sound-speed reconstruction —** Real-time per-point tissue sound speed for linear-array imaging; used to optimize B-mode time-of-flight, improving resolution and focusing.
2. **Single-focus transmit imaging —** Phase correction for focused transmit (better resolution & frame rate); derived and implemented an approximate analytic solution to the Fresnel diffraction integral; applied phase correction to phased-array (cardiac) and linear-array B-mode on real systems.
3. **PW mode auto-optimization —** One-click auto-tuning of spectral gain, velocity scale, and baseline.
4. **Development platforms —** Non-uniform medium platform; general-purpose data-exchange platform.
5. **System-level front-end —** Built the full transmit/receive algorithm chain; improved virtual-source and multi-beam beamforming for out-of-focus regions.
6. **Image tuning —** Optimized transcranial B-mode and carotid-brain integrated imaging (one of the best transcranial B-mode images on the market).

Publication

- Chao Wang, Ming Yan, Junjie Yu. “Sorted L1/L2 Minimization for Sparse Signal Recovery.” Journal of Scientific Computing. (Mathematics journals list authors alphabetically.)

Graduate Research

- Applied compressed sensing to variable selection, signal denoising, and accurate sparse recovery of high-dimensional data
- Combined optimization with deep-learning priors (Plug-and-Play, PnP) for image reconstruction

Awards & Honors

- National Mathematical Contest in Modeling for College Students — Provincial Third Prize
- Interdisciplinary Contest in Modeling (ICM/MCM) — Honorable Mention
- Comprehensive Assessment Scholarship — Third Prize
- Academic Excellence Award, Hunan Normal University
- Student Science & Technology Innovation Award

Skills

- Simulation & Imaging: proficient in k-Wave and Field II; full and accurate wave-simulation and B-mode / C-mode / PW imaging capability
- Programming: MATLAB, C++, Python
- Language: CET-6 (College English Test Band 6); regularly tracks frontier literature in ultrasound and seismic imaging

Professional Summary

- Continuously tracks state-of-the-art advances in the field and translates proven results into practice
- Active and creative thinker
- Values an open, free, and collaborative team culture built on candid discussion
- Skilled in on-time delivery and long-term project planning